



GENQUANTIS
PVT. LTD.

"As quantum computing transforms the world, make sure your career is ready for what's next."



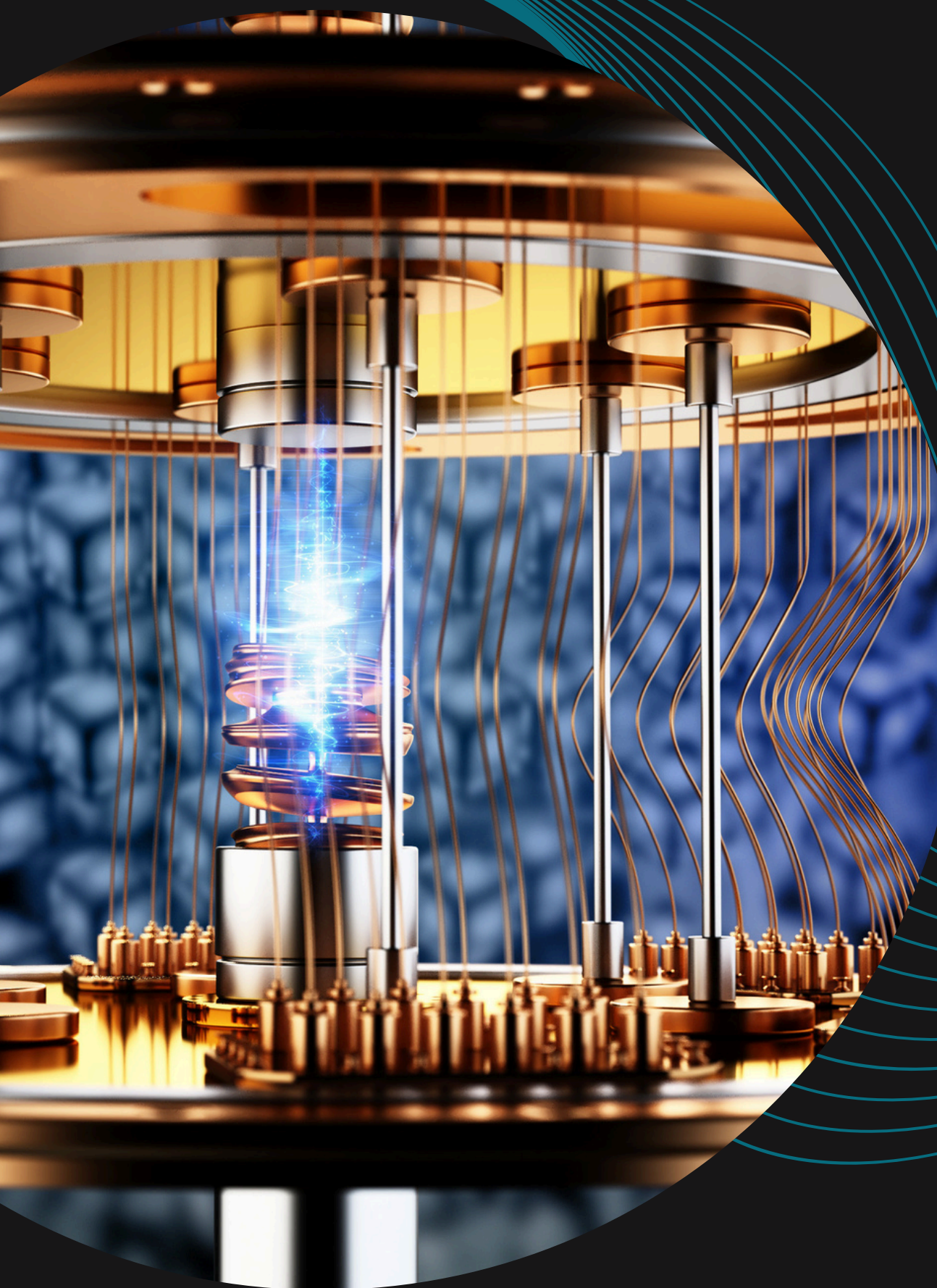
Certification in Quantum Computing

- *4 Case Studies and Projects*
- *Latest Industry Insights & Use Cases*
- *Why Quantum Matters for Everyone*
- *Real-World Applications*

Module 1: Foundations (Weeks 1-5)

MODULE 2: IMPLEMENTATION &

ALGORITHMS (WEEKS 6-10)



**REGISTER
NOW**



Eligibility Criteria:

Engineering Graduates & IT

Experienced Professionals (0-10Y exp)



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Course Snapshot:

1. Basic Course(3-6 months)

1. *Mathematical Preliminaries*

2. *Quantum Mechanics Fundamentals i*

3. *Quantum Mechanics Fundamentals ii*

4. *Introduction to qubits*

5. *Multi-qubit Systems and Entanglement*

6. *Physical Realizations & Qiskit i*

7. *Physical Realizations & Qiskit ii*

8. *Basic Quantum Algorithms I*

9. *Basic Quantum Algorithms II*

10. *Quantum Speed-up & Complexity*

11. *Quantum Fourier Transform (QFT) i*

12. *Quantum Fourier Transform (QFT) ii*

13. *Factoring Algorithm I (Shor's)*

14. *Factoring Algorithm II (Shor's)*

15. *Grover Search I*

16. *Grover Search II*

17. *Advanced Quantum Algorithms I*

18. *Advanced Quantum Algorithms II*

19. *Quantum Cryptography I*

20. *Quantum Cryptography II*

Module 1:
Foundations
(Weeks 1-5)

Module 2:
Implementation & Algorithms (Weeks 6-10)

Module 3:
Transformative Algorithms (Weeks 11-15)

Module 4:
Advanced Topics & Future Outlook
(Weeks 16-20)



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Course Snapshot:

2. Advance Course(6-12 months)

1. *Noise and Quantum Error Correction (QEC) I*
2. *Noise and Quantum Error Correction (QEC) II*
3. *Error Mitigation*
4. *Quantum Software Tools*
5. *Introduction to Pulse Programming*
6. *Variational Quantum Algorithms*
7. *Solving Optimization Problems (Part 1)*
8. *Solving Optimization Problems (Part 2)*
9. *Quantum Machine Learning (QML) & Quantum Communication*
10. *Quantum Communication Protocols & Advanced Topics*

Module 5:
**Noise, Error,
and Software**
(Weeks 21-35)

Module 6:
**Modern
Applications
and Future
Trends**
(Weeks 36-48)





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PVLE.HD.

Course Comparison with peers

Feature	QuantXAI Our Combined Course Offering	IBM Quantum Learning	Google Quantum AI / Cirq	IIT Madras / NPTEL
Primary Focus	Holistic, Applied Full-Stack (Theory, Qiskit mastery, Advanced Applications, & Hardware- Specifics)	Qiskit Ecosystem & industry standards; broad foundation.	Hardware & Error Correction focus; Cirq/TensorFlow Quantum integration.	Deep Academic Rigor; strong theoretical foundation; NPTEL often self- paced/worksho p.
Coding Platform	Qiskit (Basic & Advanced), D- Wave/Quantum Annealers (Advanced)	Qiskit (Primary)	Cirq, TensorFlow Quantum, Stim, Crumble	Qiskit (Common in NPTEL/online courses)
Advanced Topics	QML, VQE, QUBO/Optimization (Part 1 & 2), QEC/Error Mitigation (Specific Modules), Advanced Cryptography (BB84).	Strong in QEC, QML, VQE (via specialized courses).	Exceptionally strong and detailed focus on Quantum Error Correction (QEC) and Quantum Hardware (Theory/Simulati on).	Strong foundation in Theory/Algorith ms (Shor's, Grover's); sometimes covers VQE and QML.
Hardware Access/ Low-Level Programm ing	Dedicated Module on Pulse Programming (Module 9 - Advanced) and D- Wave/Annealer Focus (Module 10 & 11 - Advanced).	Access to real IBM Quantum Systems; Qiskit Runtime optimization.	Access via Google Cloud Platform (GCP); focuses on hardware development/th eory.	Cloud access often provided via IBMQ Platform partnership.



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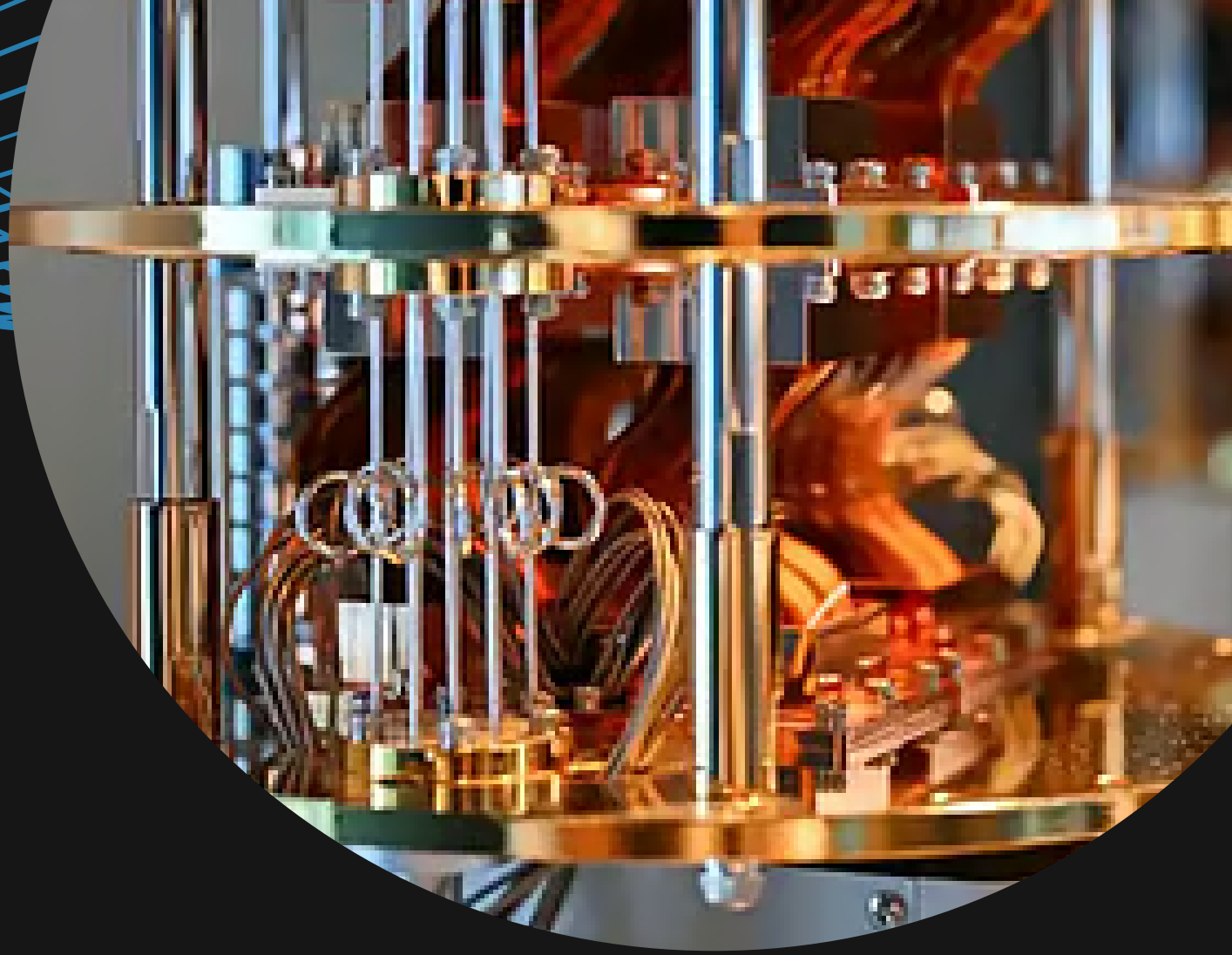
Projects Consist:

1. Advanced Projects

- *Qubit State Visualization and Measurement Simulation*
- *Implementation and Verification of the Bell State Circuit*
- *Quantum Teleportation Protocol Simulation using Qiskit*
- *Comparative Analysis of Qiskit Simulator vs. IBMQ Hardware Performance*
- *Coding and Testing the Deutsch-Jozsa and Bernstein-Vazirani Algorithms*
- *Practical Implementation of the Quantum Phase Estimation Algorithm*
- *Grover's Search Algorithm: Oracle Design and Demonstration*
- *Simulation of the BB84 Quantum Key Distribution Protocol*

2. Super Advance Projects

- *Variational Quantum Classifier (VQC) for Data Classification (e.g., Iris Dataset)*
- *Solving the Max-Cut Combinatorial Optimization Problem using VQE/QAOA*
- *QUBO Formulation and Solution for Advanced Optimization Problems*
- *Noise Mitigation: Implementing and Benchmarking Readout Error Correction*
- *Quantum Software Engineering: Transpilation and Circuit Cutting Optimization*
- *Full Implementation of Super-dense Coding on Real Quantum Hardware*
- *Low-Level Quantum Control: Demonstrating Rabi Oscillations via Pulse Programming*
- *Advanced Search: Generalizing Amplitude Amplification Techniques*
- *Quantum Finance Application: Portfolio Optimization using QUBO*





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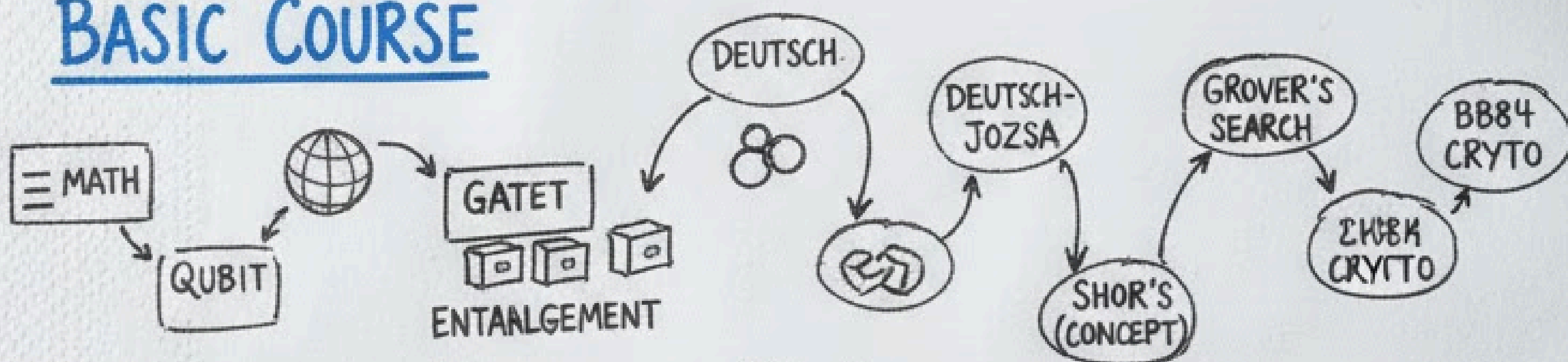
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Roadmap of Quantum



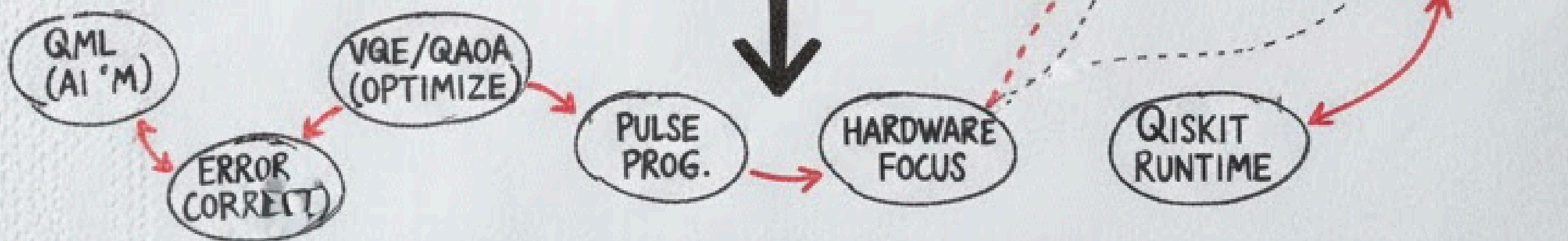
QUANTUM LEARNING JOURNEY

BASIC COURSE



ADVANCED COURSE

YOUR
FULL-STACK
PROGRAM!



READY FOR QUANTUM FUTURE!

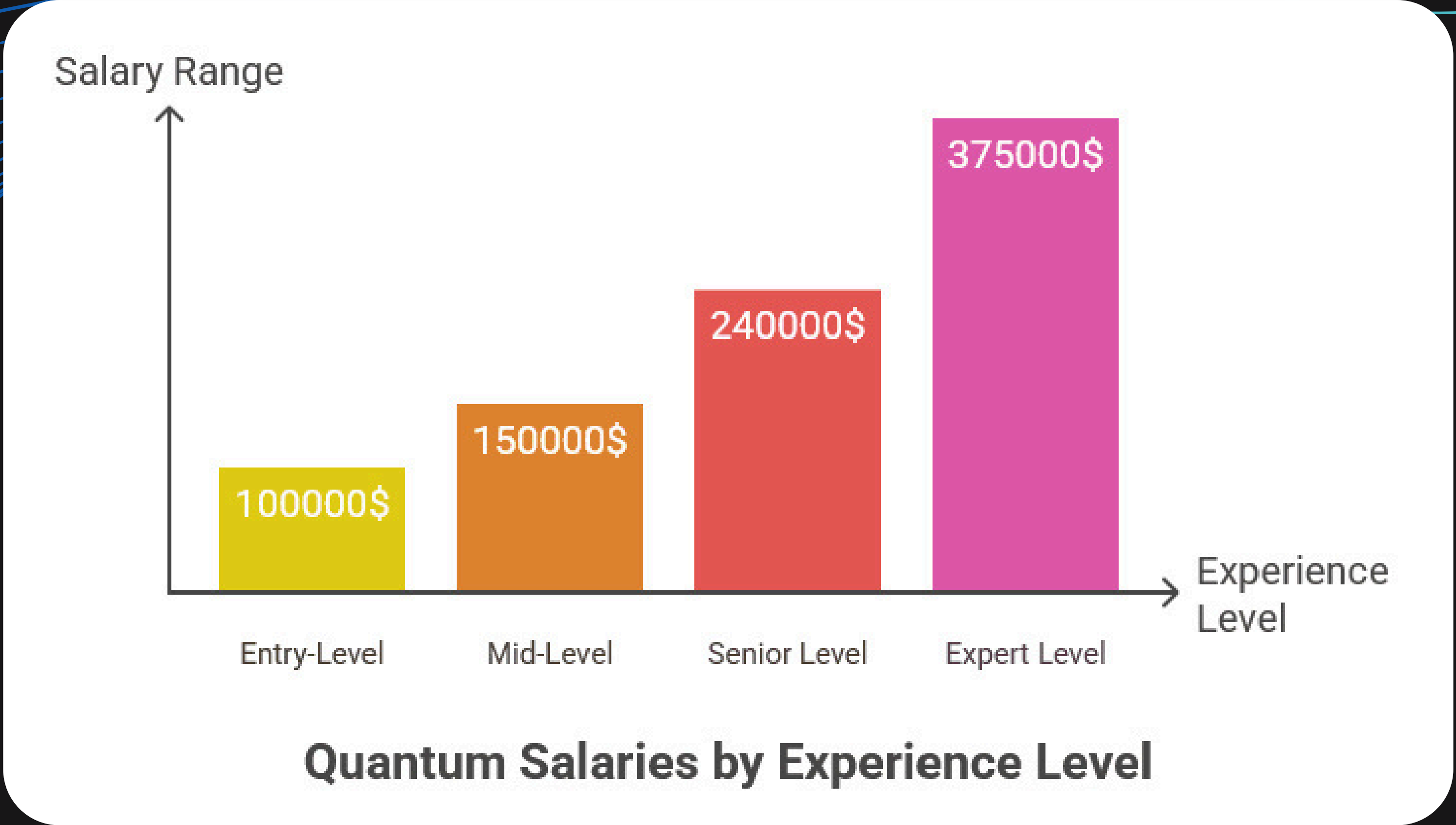


Salary Standards



Quantum Software Engineer	Salary: \$120,000 - \$160,000
Quantum Research Scientist	Salary: \$130,000 - \$180,000
Quantum Hardware Engineer	Salary: \$140,000 - \$190,000
Quantum Algorithm Developer	Salary: \$125,000 - \$175,000
Quantum Cryptography Specialist	Salary: \$130,000 - \$170,000
Quantum Information Theorist	Salary: \$120,000 - \$160,000
Quantum Systems Engineer	Salary: \$135,000 - \$185,000
Quantum Applications Scientist	Salary: \$125,000 - \$165,000
Quantum Machine Learning Researcher	Salary: \$130,000 - \$180,000
Quantum Consultant	Salary: \$150,000 - \$200,000

Quantum Salaries by Experience Level

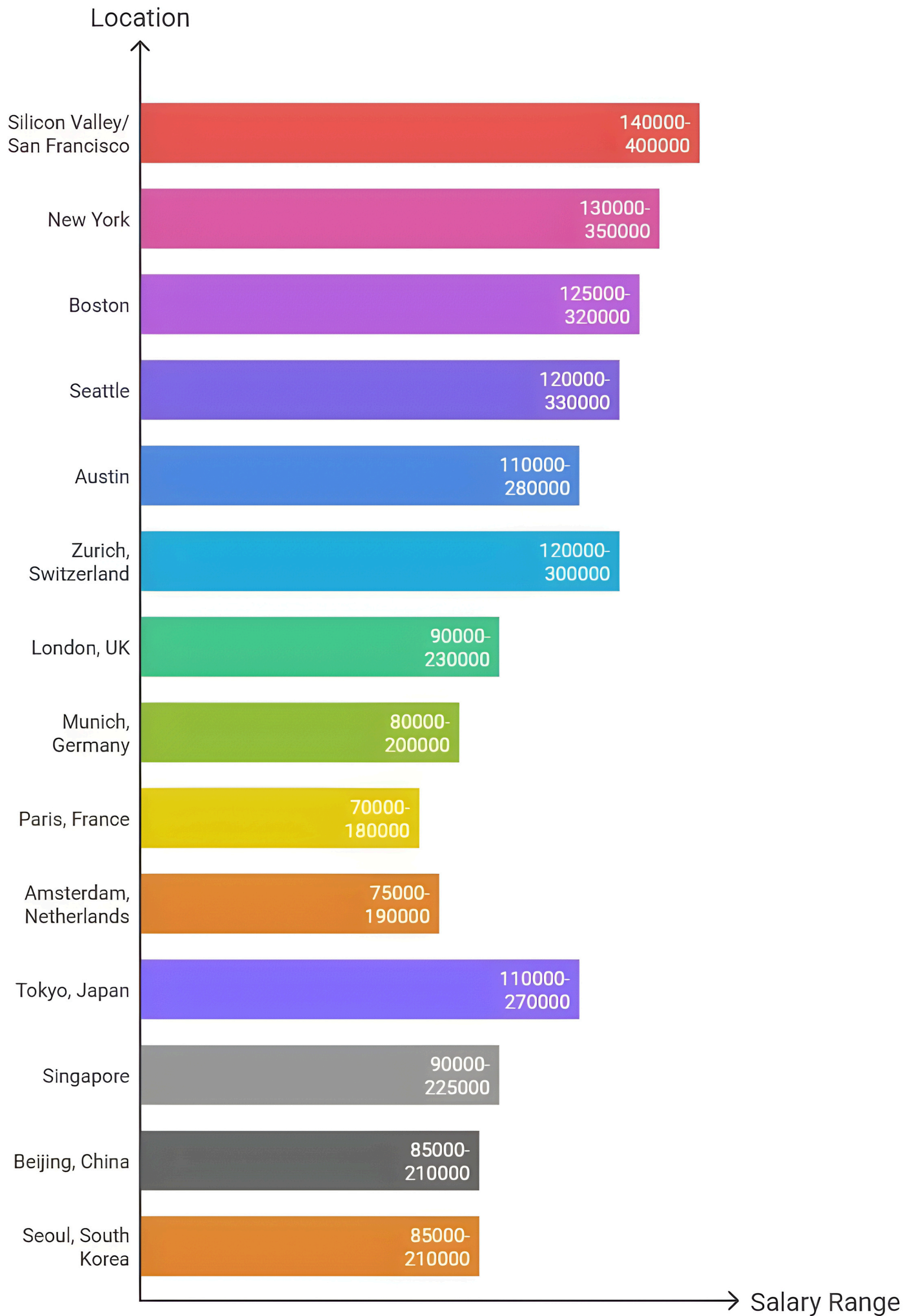


High Impact Quantum Bonus Skills

Skill	Importance	Salary Boost Factor	Bonus Skills
 Qiskit (IBM Quantum)	Widely used SDK	Design and simulate circuits	Qiskit Aer, Qiskit Runtime
 Quantum Hardware Familiarity	Understanding qubit types	High-paying roles in hardware	Cryogenics, microwave engineering
 Python with Quantum Libraries	De facto language for quantum dev	Rare cross-domain developers	JAX, PyTorch, TensorFlow Quantum
 Error Correction & Fault-Tolerant	Crucial for scalable systems	High-paying R&D roles	Qubit overhead, error thresholds
 Quantum Algorithms	Design and optimize algorithms	Essential for finance, chemistry	QAOA, VQE

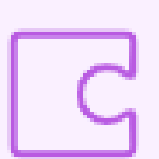


Quantum Global Jobs & Salaries by Location





Quantum Functional Roles in India

Company/Institution	Role Title	Salary Range (INR)	Primary Focus
 IBM India	Quantum Software Engineer	₹10 LPA – ₹20 LPA	Qiskit, Cloud, Applications
 TCS Research & Innovation	Research Scientist	₹18 LPA – ₹35 LPA	Optimization, QML, Research
 Microsoft Research India	Quantum Research Scientist	₹20 LPA – ₹40 LPA+	Advanced QML, Software, QEC
 Accenture / Deloitte	Quantum Analyst/Specialist	₹12 LPA – ₹25 LPA	Strategy, Studies, Applications
 C-DAC (National Labs)	Scientist/Engineer	₹8 LPA – ₹18 LPA	Government, QEC, Simulation, Algorithms
 Indian Startups	Algorithm Developer	₹8 LPA – ₹16 LPA	Optimization, Industry Applications

Let's Build Your Future



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Have questions about our courses, certifications,
or career guidance?

Our team is here to help you choose the right
learning path and get started on your Quantum
journey.

Our Phone:

7036955133



Our Website:

www.quant-xai.com



Our Email:

academy@genquantis.com



**Enroll
Now**

